Soham Patil

soham.patil@research.iiit.ac.in • soham2560.github.io • github.com/soham2560 • linkedin.com/in/sohampatil2560 • Google Scholar

Education

IIIT Hyderabad, Robotics Research Center — Jul 2024 – Present
MS by Research, Computer Science • CGPA: 10.0/10 — *Hyderabad, India*
Advisor: Prof. Spandan Roy

MIT World Peace University (MIT-WPU) — Jul 2019 – Jul 2023
B.Tech, Computer Science • CGPA: 8.55/10 — *Pune, India*

Publications

NoContactNoWorries: Estimating Contact through Vision and Proprioception for In-Hand Dexterous Manipulation [website] — *IROS 2026*
S. Patil, A. Das, S. Bhosale, S. Roy

AnyViewDex: View-Invariant Dexterous Manipulation from RGB Observations [website] [arXiv] — *under review at ICRA 2027*
S. Patil*, O. S. Gunjal*, S. Bhosale, A. Chavare, R. S. Hora, S. Roy

Aerograb: A Unified Framework for Aerial Grasping in Cluttered Environments [arXiv] — *IROS 2026*
S. P. Singh, N. S. Nair, S. Ujjawal, S. Mishra, **S. Patil**, R. D. Yadav, S. Roy

* Equal contribution.

Experience

Google Summer of Code @ ros2-control [code] — 2025
Open-Source Contributor — *Remote*

- Built and merged upstream hardware-diagnostics support for *ros2_control*: a standardized health/fault-reporting API, new ROS messages (HardwareStatus, HardwareDeviceStatus), a unified Actuator/Sensor/System interface, and default diagnostic nodes with CLI tooling. Mentors: Dr. Bence Magyar, Sai Kishor Kothakota.

Robotics Research Center, IIIT Hyderabad — Feb – Jul 2024
Research Intern — *Hyderabad, India*

- Built an in-house collaborative robot arm and an end-to-end pipeline for testing custom controllers on hardware, supporting research on adaptive/robust control.

Honors & Awards

DD Robocon, India's national stage of ABU Robocon (the Asia-Pacific university robotics championship) — 2020–23

- **Winner** (2020), represented India at the internationals; ranked **2nd** (2023) and **4th** (2022) nationally among 150+ teams.
- Led a team designing, building complete competition robots from scratch each season across mechanical design, custom electronics, firmware, motor control, and autonomy; built an in-house integrated actuator matching commercial specs at ∼half the cost, a wheeled-odometry module (<4% err), and ROS 2/Nav2 autonomy on register-level TI/STM32 FW.

MATLAB Winners Circle: omni-drive robot modeling and control in MATLAB/Simulink — 2020

Teaching

Teaching Assistant, IIIT Hyderabad: Mobile Robotics (CS7.503), Robotics: Planning & Navigation (EC4.403), Systems Thinking (EC5.202), Perception & Planning for Aerial Vehicles (NPTEL) — 2024–present
Instructor, RRC Robotics Summer School: ROS deployment (teleoperation, autonomous navigation) — 2025

Service

Session Chair, IROS 2026: *Imitation and Demonstration Learning for Robot Skills* — 2026
Session Co-Chair, IROS 2026: *Policies and Retargeting for Bimanual Manipulation* — 2026

Technical Skills

Programming	C, C++, Python
Machine Learning	PyTorch, NumPy, OpenCV, Isaac Lab, MuJoCo
Robotics	ROS 2, ros2_control, Nav2, MoveIt2, Gazebo, RViz, LEAP Hand, xArm7
Embedded	STM32, TM4C, AVR MCUs and peripherals (CAN, PWM)
Tools & DevOps	Linux, Git, Docker, Ansible, GitHub Actions, CI/CD